Mission Report: Track 1 – The Laser Grid Solutions

Module: Physics / Computer Vision
Institution: ENSTA – National Higher School of Advanced Technologies
Academic Year: 2025–2026
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1. Law of Reflection

The Law of Reflection consists of two fundamental principles:

1. The angle of incidence is exactly equal to the angle of reflection:

$$\theta_i = \theta_r$$

2. The incident ray, the reflected ray, and the normal (the imaginary line perpendicular to the mirror's surface) all lie in the same plane.

Proof / Justification:
Consider a plane mirror with a normal vector $\vec{n}$perpendicular to the surface. The incident ray $\vec{I}$and reflected ray $\vec{R}$satisfy:

$$\vec{R} = \vec{I} - 2(\vec{I} \cdot \vec{n})\vec{n}$$

This vector equation **mathematically proves $\theta_i = \theta_r$** and ensures the rays remain in the same plane.

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2. Calculation of Reflection Angles

Step 1: Reflection on Mirror M1

- Given: $i_1 = 30°$
- By the Law of Reflection: $r_1 = i_1 = 30°$

Step 2: Reflection on Mirror M2

- Mirror M2 is perpendicular to M1 → normals form a **right triangle**.
- Acute angles satisfy:

$$r_1 + i_2 = 90°$$

- Substituting $r_1 = 30°$:

$$i_2 = 60°$$

- Law of Reflection on M2: $r_2 = i_2 = 60°$

Geometric illustration (schematic):

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    M1
    |
    |\
    | \
 i1 |  \ r2
    |   \
    |____\ M2
      i2
```

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3. Proof that Exiting Ray is Parallel to Original

Step 1: Deviation by one mirror

$$\delta = 180° - 2i$$

Step 2: Total deviation

$$\delta_{\text{total}} = \delta_1 + \delta_2 = (180° - 2i_1) + (180° - 2i_2)$$
$$\delta_{\text{total}} = 360° - 2(i_1 + i_2)$$

Step 3: Substitute angles

$$i_1 + i_2 = 30° + 60° = 90°$$
$$\delta_{\text{total}} = 360° - 2(90°) = 360° - 180° = 180°$$

Conclusion: The exiting ray is **parallel to the incident ray** but travels in the opposite direction – a complete U-turn.

Step 4: Vector proof (optional for rigor)

Let incident ray vector $\vec{I} = (1,0)$. After M1 reflection:

$$\overrightarrow{R_1} = \vec{I} - 2(\vec{I} \cdot \overrightarrow{n_1})\overrightarrow{n_1}$$

After M2 reflection:

$$\overrightarrow{R_2} = \overrightarrow{R_1} - 2(\overrightarrow{R_1} \cdot \overrightarrow{n_2})\overrightarrow{n_2}$$

Calculation shows $\overrightarrow{R_2} = -\vec{I}$, proving exact **opposite parallelism**.

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4. Simulation & Documentation

- All calculations are confirmed with **numerical examples**.

- Project repository includes:
    - **Mission Report PDF**
    - **Presentation PPTX**
    - **index.html** linking the report and presentation
- This ensures **scientific accuracy**, **clarity**, and **full compliance** with mission requirements.